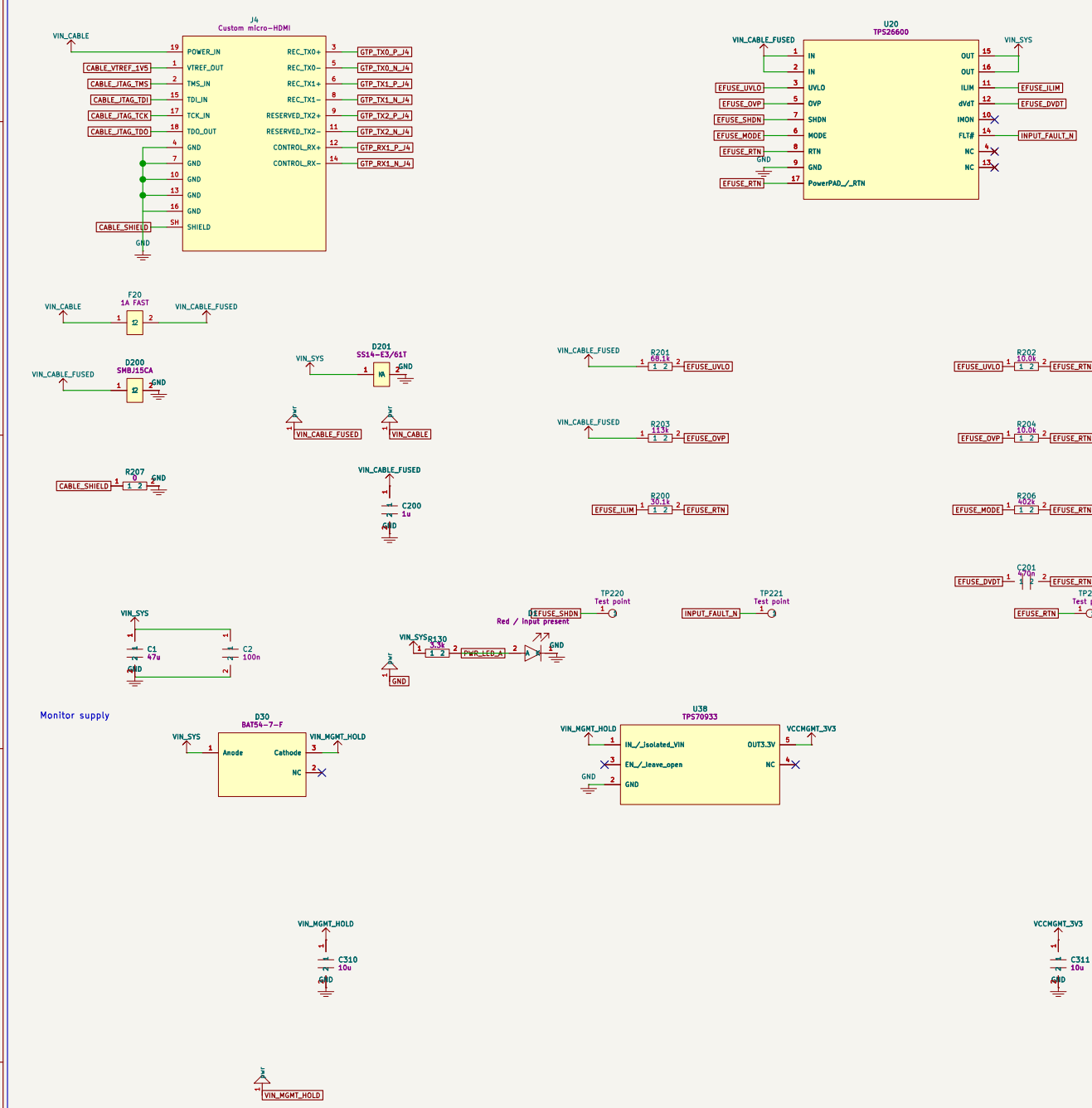
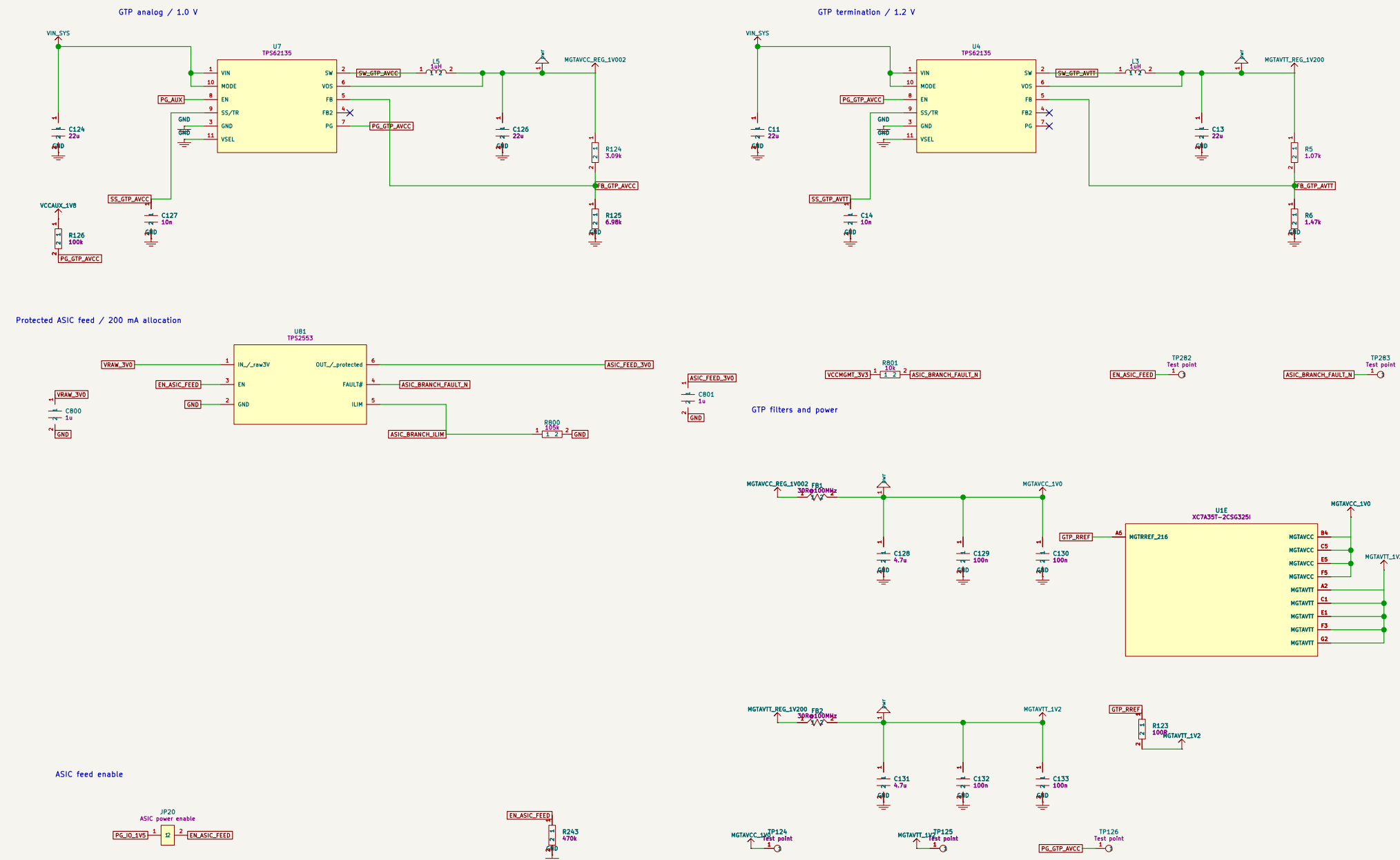


1 Cable input and protection



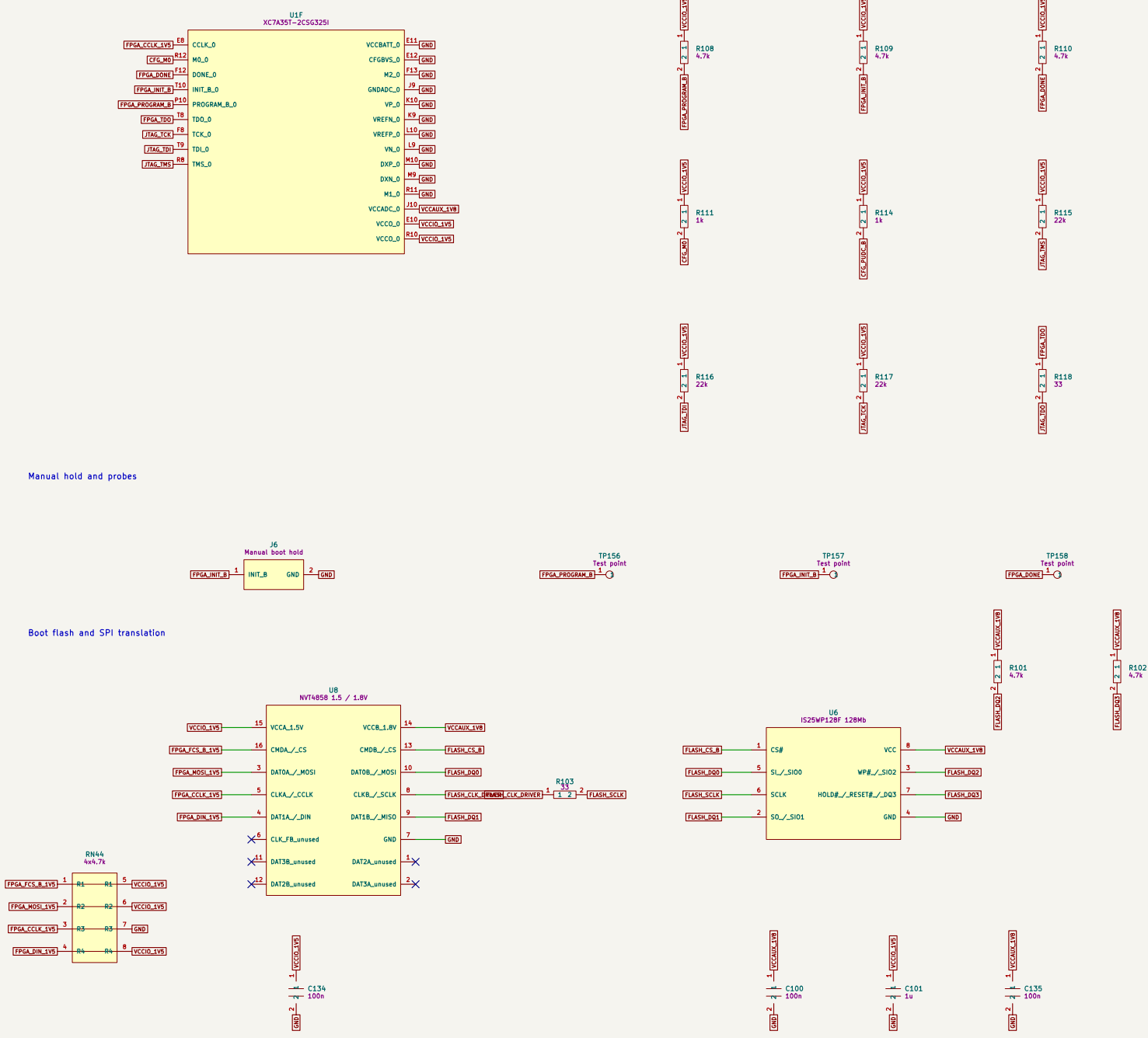
Custom cable power, JTAG and serial data.
FPGA_PFN (pin 17) must stay separate from GND.

2 ASIC feed and serial-link supplies



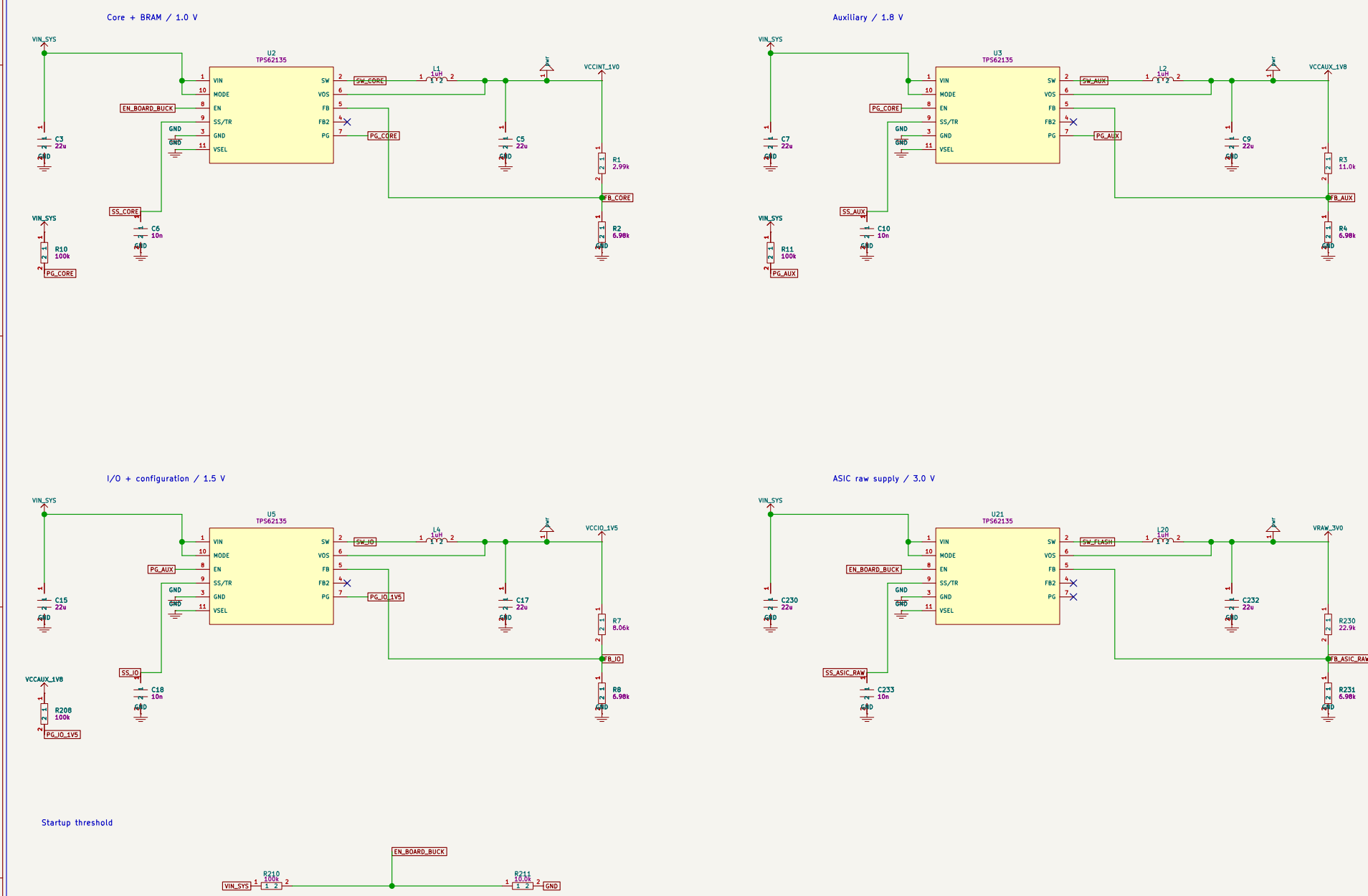
Filtered 1.0 V / 1.2 V rails supply the GTP block.
UBI limits the ASIC feed enable follows I/O power-gate.

3 FPGA configuration and boot flash



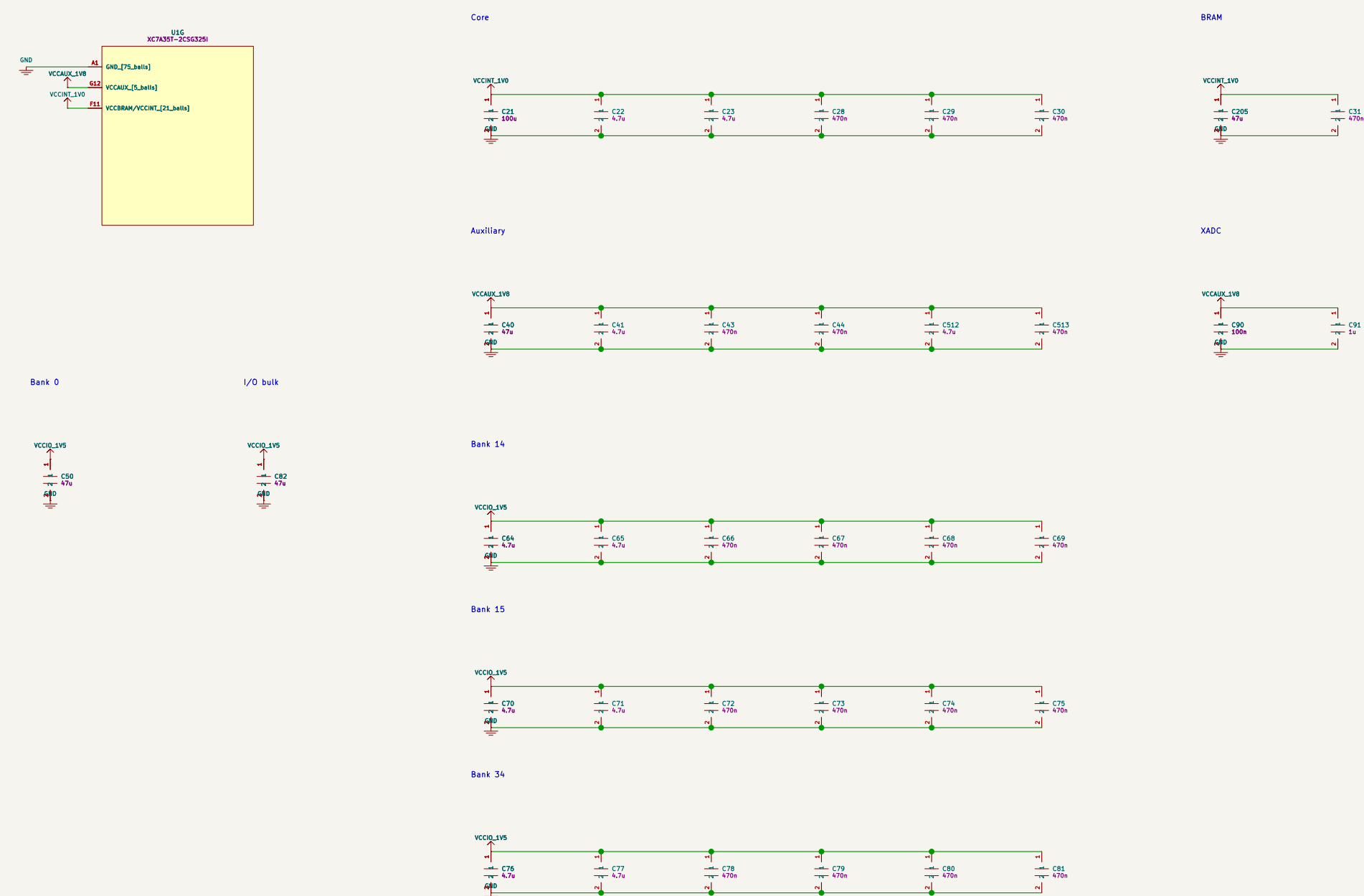
W70000 pins 1.0 V FPGA to 1.0 V SPI at flash.
J0 holds SWL0 low for manual recovery.

4 FPGA and ASIC converters



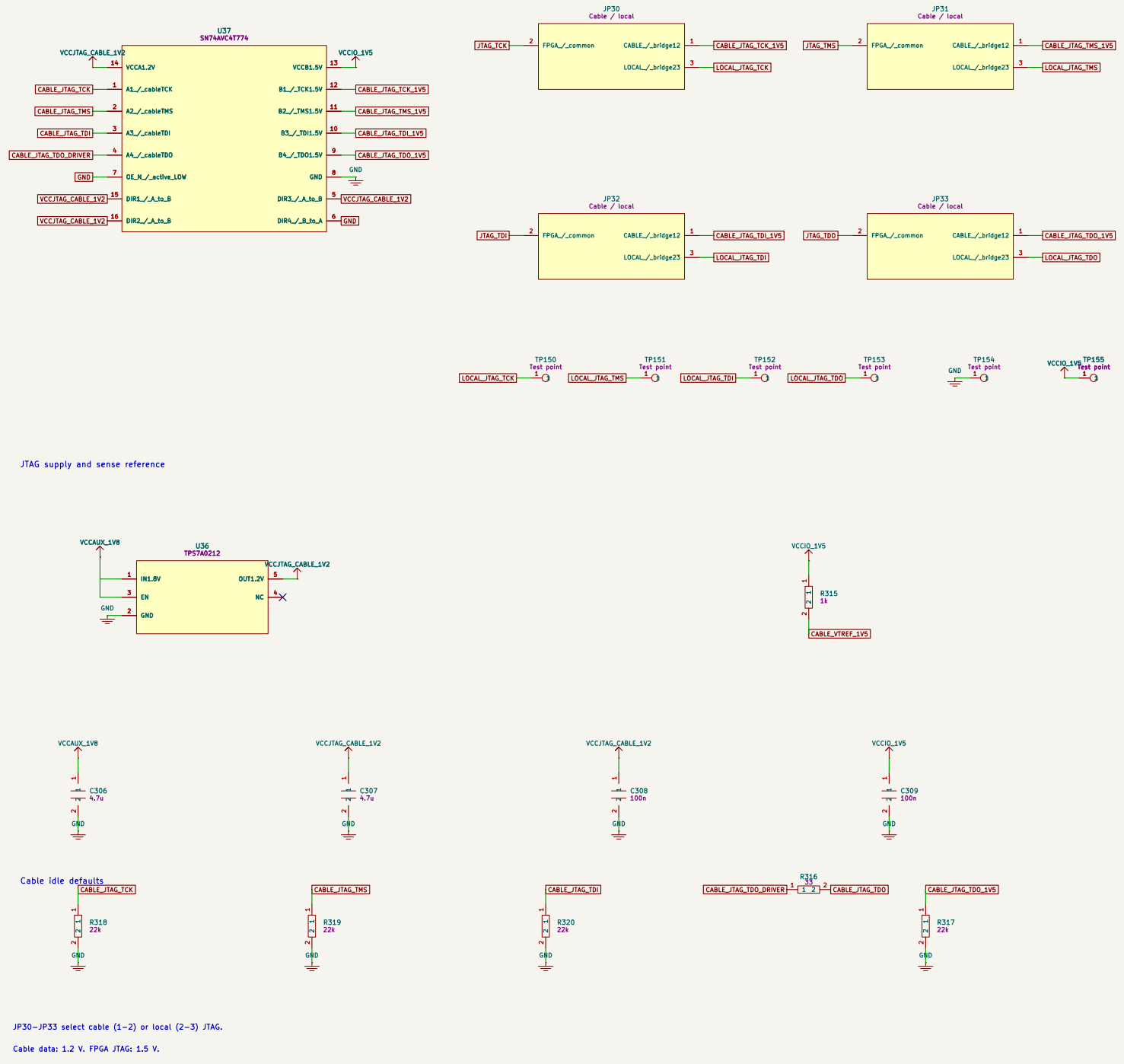
1.0 V core, 1.0 V auxiliary and 1.0 V I/O.
1.0 V feeds the ASIC branch. Boot flash uses 1.0 V.

5 FPGA power and bypass



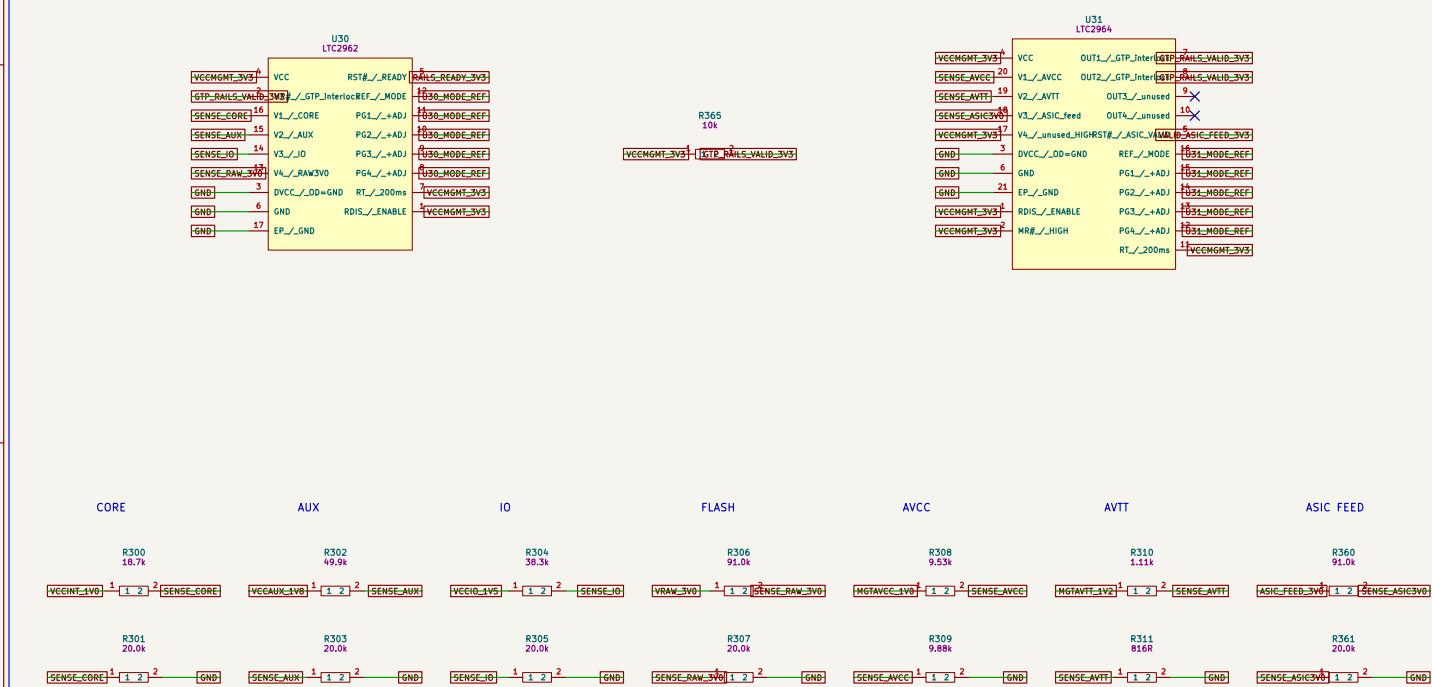
Local bypass for FPGA power pins and I/O banks.

6 Programming and recovery



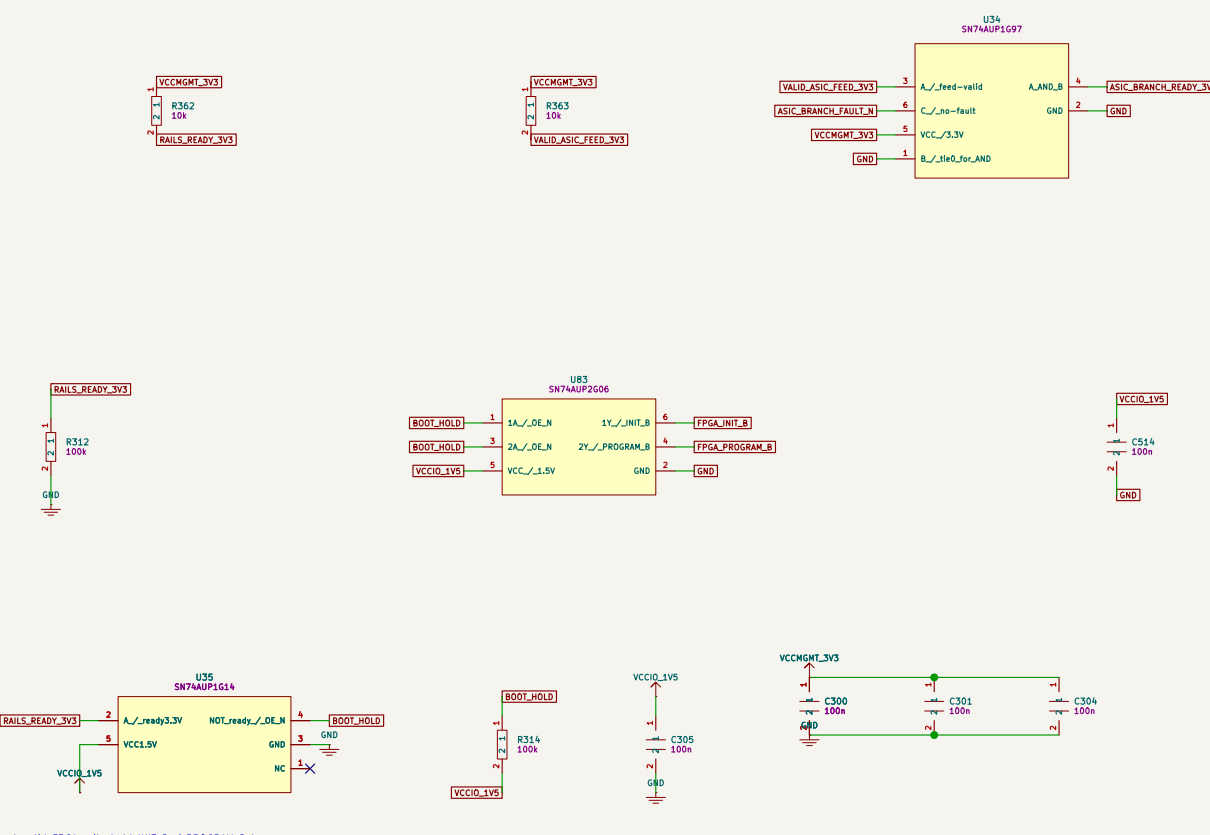
JTAG supply and sense reference.
Cable line delay.
JTAG supply and sense reference.

7 Rail monitors



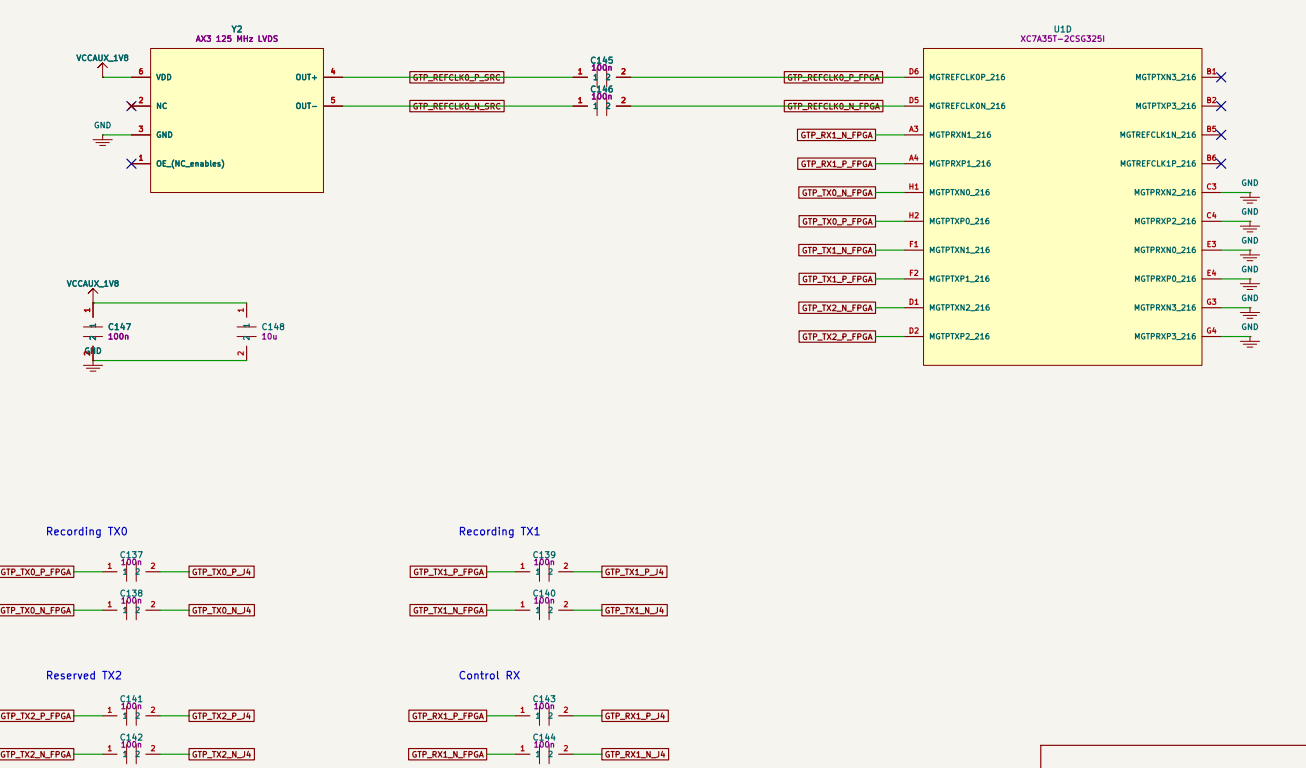
Recovery hold 100-200 ms.
Watch reset at FPGA brownout response needs validation.

8 Boot hold and release



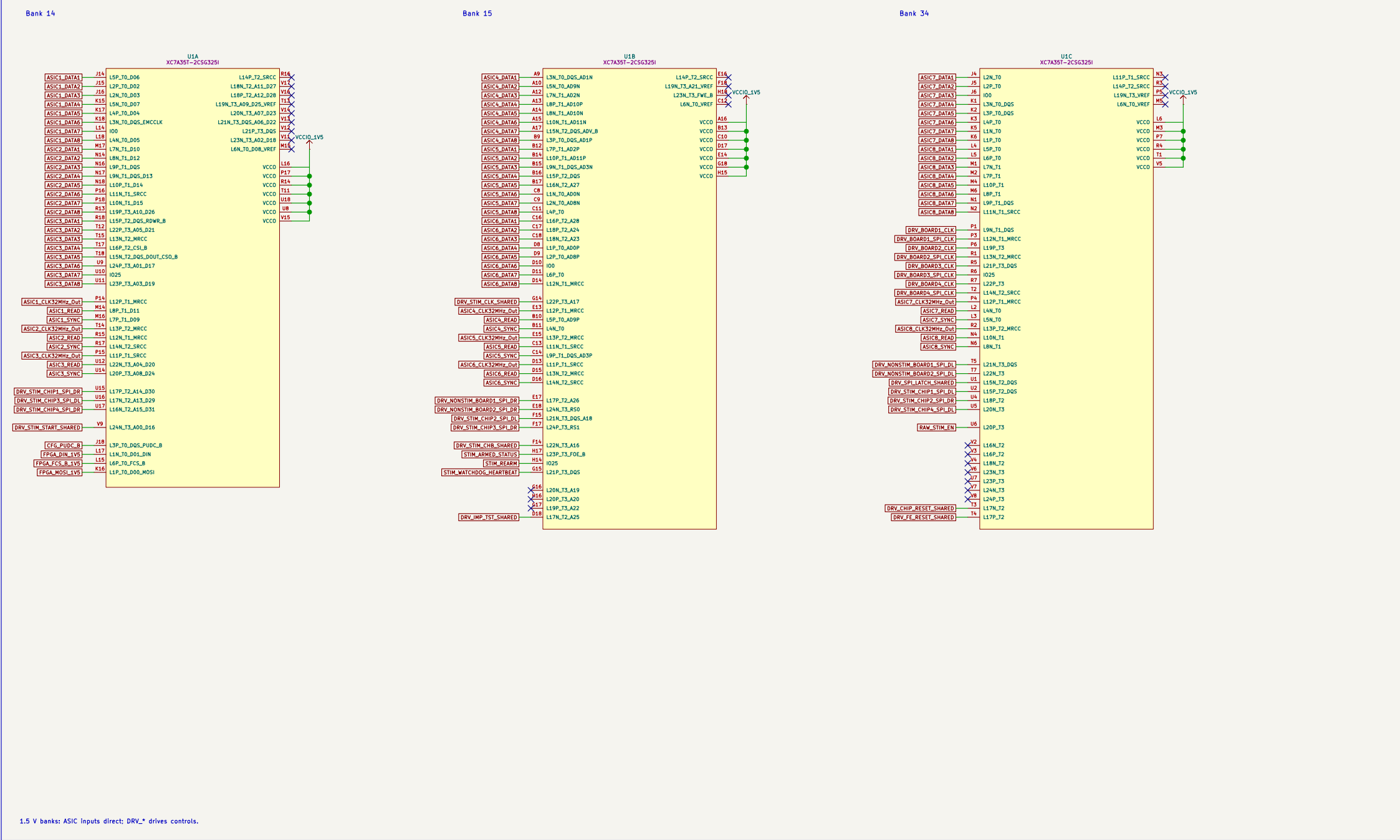
Watch FPGA pins hold INT0 / PROGRAM low.
ASIC feed validity is queried separately.

9 Reference clock and serial link

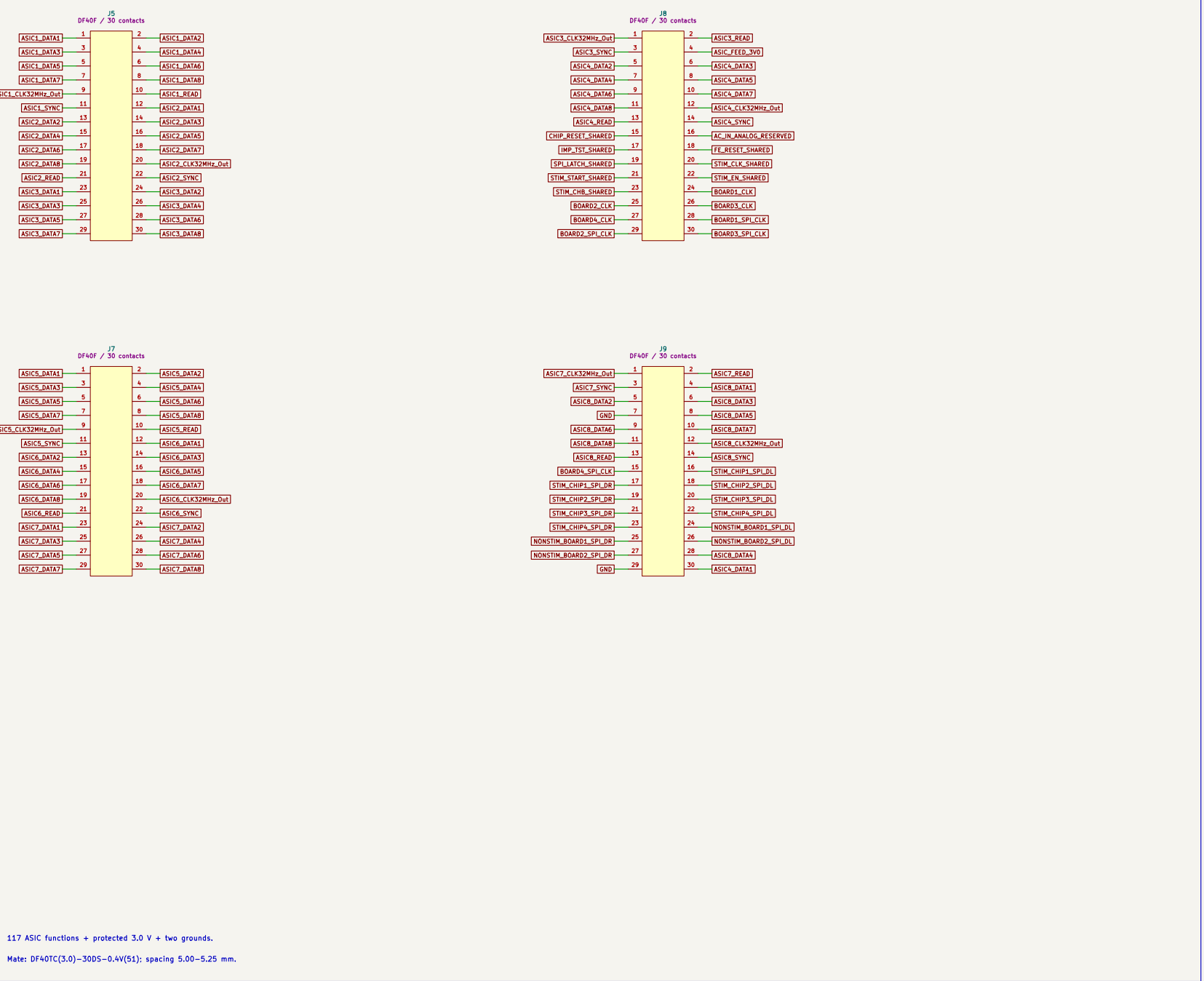


120 MHz reference for the serial link.
ASIC output uses each chip's reference clock.

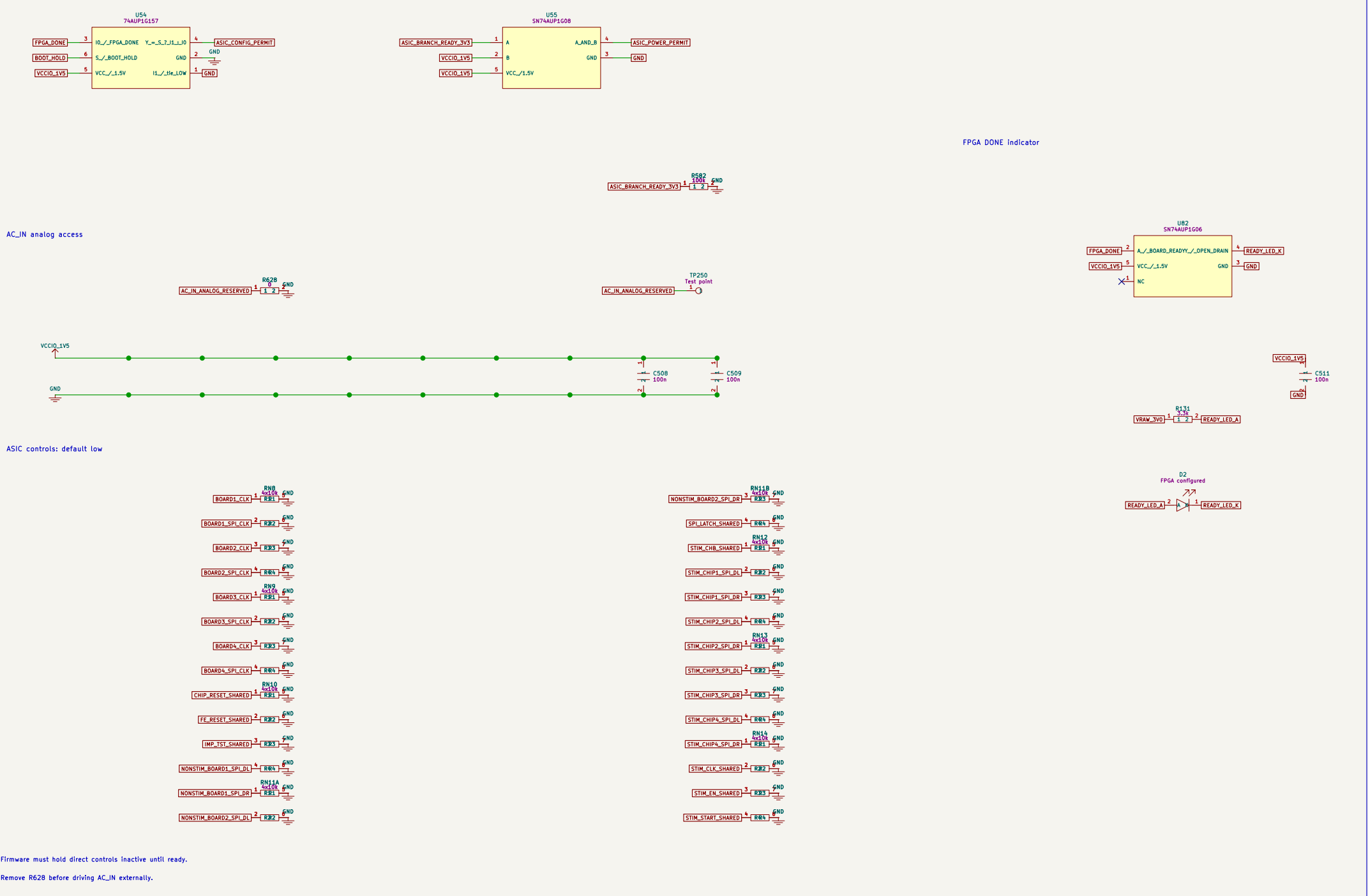
1 FPGA signal banks



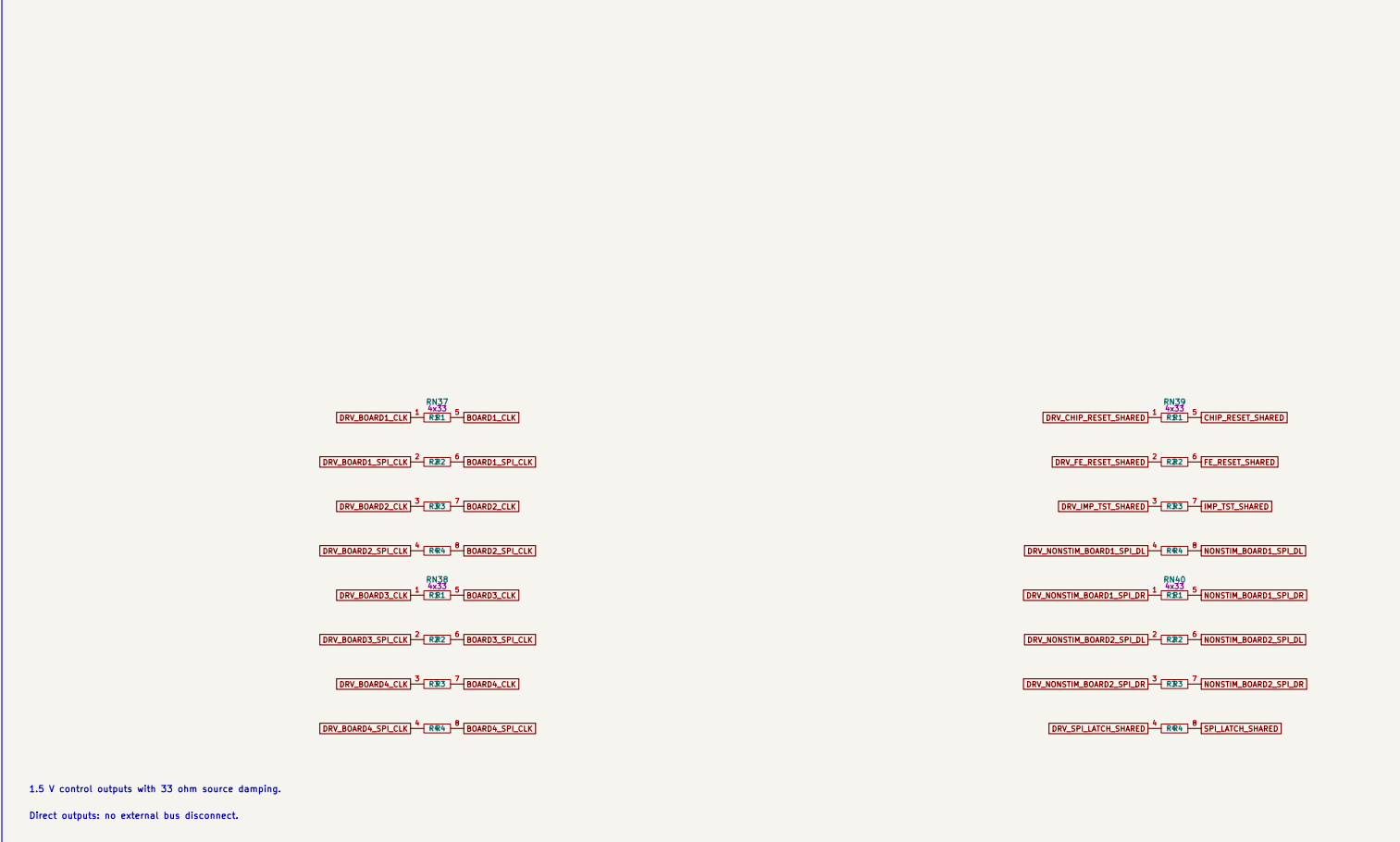
2 ASIC mezzanine connectors



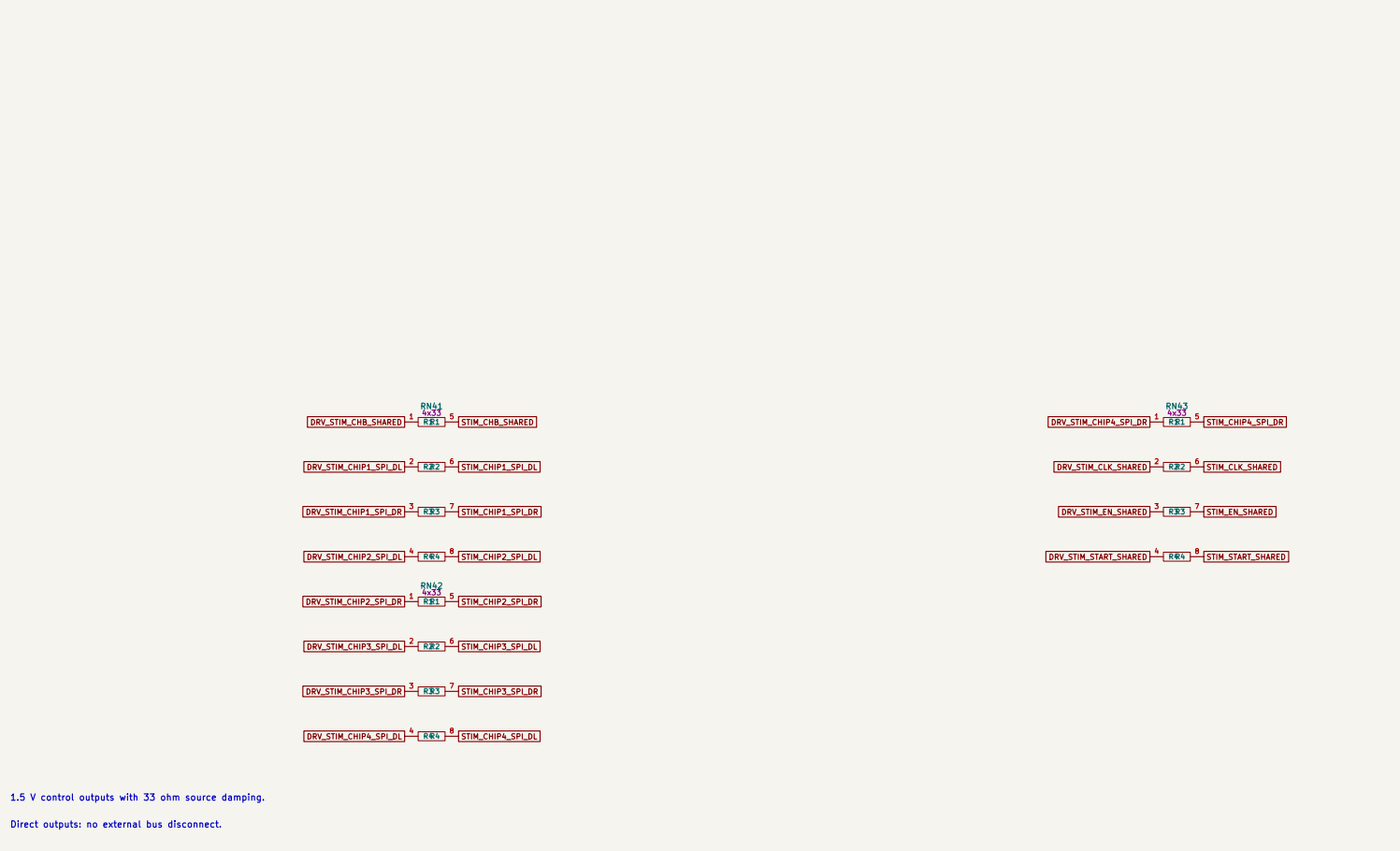
3 Enable, defaults and status



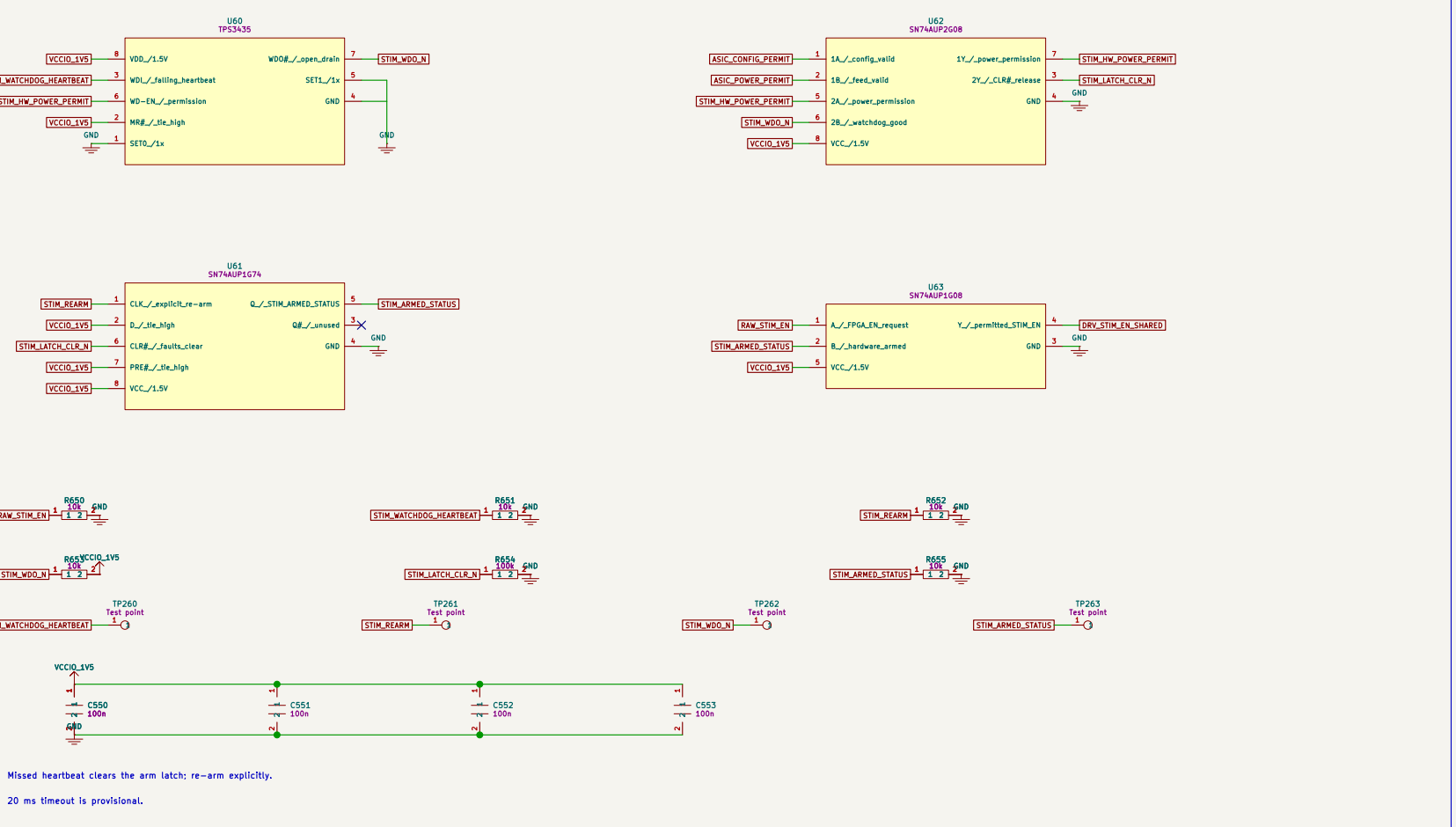
4 Direct controls and damping 1



5 Direct controls and damping 2

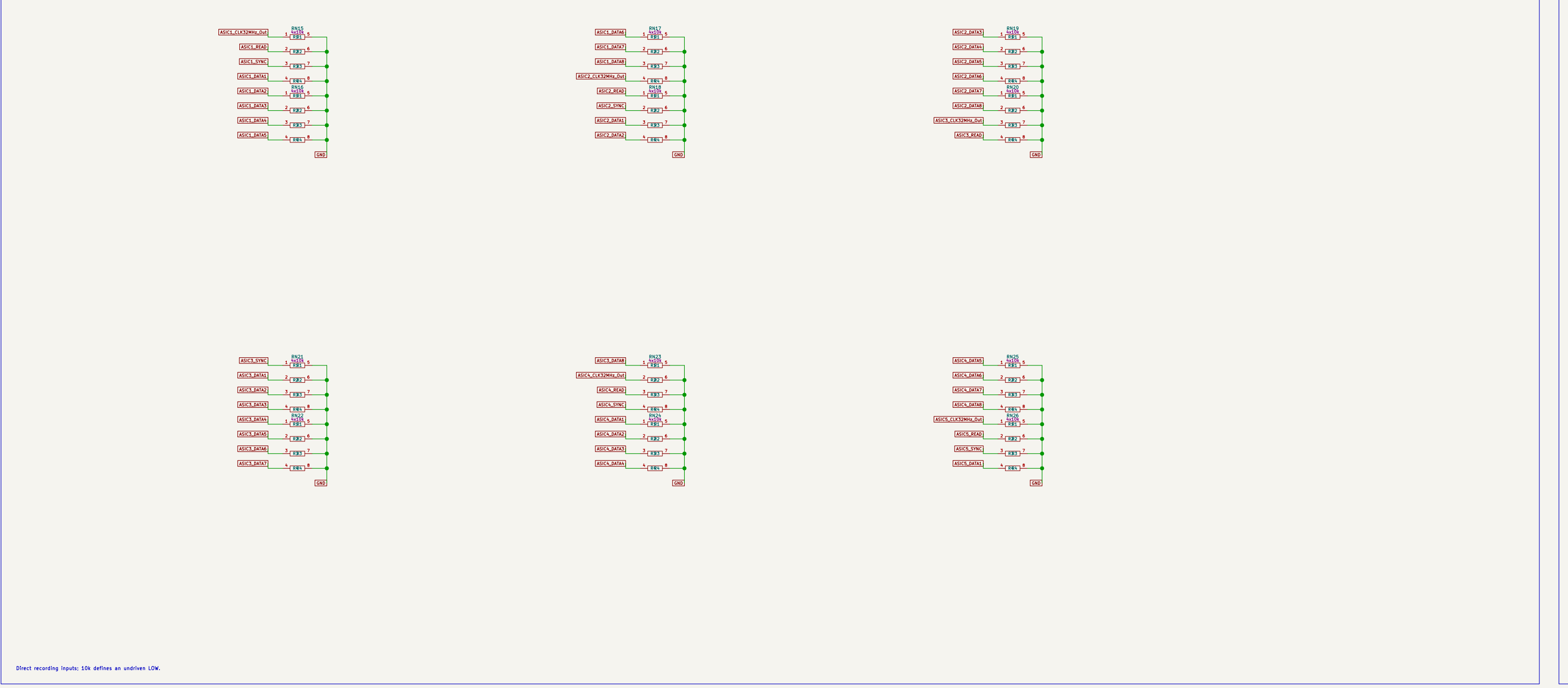


6 STIMLEN watchdog and re-arm



D2 indicates FPGA configuration ok.
C1X2MuxSel is a historical net name.
Selected clock pins: 125 MHz / 6 = 31.25 MHz.
ASIC timing still needs validation.

7 Direct input defaults - channels 1-48



8 Direct input defaults - channels 49-88

